Comment on "Ergotropy and entanglement in critical spin chains"

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The occupation-alignment expression in Eq. (34) of B. Mula et al., Phys. Rev. B **107**, 075116 (2023), minimizes the subsystem energy under number-preserving Gaussian orbital rotations. It need not give the minimum over all subsystem unitaries. For the unchanged half-filled ground state of the homogeneous open fermion chain, we establish the first failure among lengths $N = 4m$ at $N = 12$. A swap between two physical two-particle states lowers the aligned energy while preserving particle number and parity. Rational interval calculations enclose the full additional extractable work between $1.356396207356 \times 10^{-6} J$ and $1.356396207357 \times 10^{-6} J$. We give a quartic extraction gate and finite comparisons through $N = 24$. The corrections amount to about 15–31 parts per million of the aligned bound energy. These finite results leave the proposed thermodynamic entropy relation undecided.

I. THE OPTIMIZATION BEING CORRECTED

Subsystem ergotropy distinguishes the energy removable by a unitary from the energy remaining above the isolated subsystem ground state. Reference [1] defines the passive energy by ordering the complete many-body spectra in Eqs. (1)–(4), but identifies it with an occupation-aligned one-body sum in Eq. (34). The distinction matters even for its specified free-fermion ground-state marginal. Ordering individual occupations against one-particle energies need not order their many-body products against subset energies.

The general distinction between unrestricted and Gaussian work extraction is established prior work [2]. Population swaps are also standard. The result here concerns the actual marginal of the chain in Ref. [1], without adjustable mode temperatures or a replacement state. The nearby long-range Kitaev study [3] treats pairing, chemical potential, and asymptotic entanglement spectra; we make no correction claim about that model. Our counterexample concerns the exact finite optimization in Eq. (34) of Ref. [1]. It does not settle the separate large-N bound-energy prediction in that reference.

II. EXACT INPUT AND TWO OPERATION SETS

Let $N = 4m$, $m \geq 1$, and $\ell = N/2$. With canonical fermions and $J > 0$, take the open Hamiltonian

$$H_N = -J \sum_{j=1}^{N-1} (c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j). \tag{1}$$

There is no chemical potential, pairing, interaction, or bath. The empty vacuum has zero energy. Let ρ_A be the marginal of the unique half-filled ground state on $A = \{1, \ldots, \ell\}$, and let H_A retain only the internal links of A after isolation. Work below is the decrease of $\mathrm{Tr}(\rho_A H_A)$; it does not include the cost of cutting the chain or implementing a control.

Solving the open one-body difference equation, with vanishing wave function at sites 0 and $N+1$, gives sine modes and energies $-2J\cos[k\pi/(N+1)]$. Filling its $N/2$ negative energies gives exactly

$$(C_A)_{ij} = \frac{2}{N+1} \sum_{k=1}^{N/2} \sin\frac{ik\pi}{N+1} \sin\frac{jk\pi}{N+1}, \tag{2}$$

where $(C_A)_{ij} = \langle c_i^\dagger c_j \rangle$, $1 \leq i, j \leq \ell$. Let $\nu_1 \geq \cdots \geq \nu_\ell$ be its eigenvalues. The physical energies of H_A, in ascending order, are

$$\varepsilon_j = -2J \cos\frac{j\pi}{\ell+1}, \qquad E_{\mathrm{al}} = \sum_{j=1}^{\ell} \nu_j \varepsilon_j. \tag{3}$$

This pairs descending occupations with ascending physical energies, giving the source expression its favorable convention. Rephasing sites or reversing one-body labels is not the counterexample.

The marginal is a number-conserving Gaussian state. In its natural orbitals, Wick factorization fixes all occupation moments and therefore the density matrix as a product of Bernoulli mode densities. In particular, all its 2^ℓ eigenvalues are

$$q(\mathbf{n}) = \prod_{j=1}^{\ell} \nu_j^{n_j} (1-\nu_j)^{1-n_j}, \quad \mathbf{n} \in \{0,1\}^\ell. \tag{4}$$

Let V contain real natural orbitals in descending occupation order, and $\Phi_{ij} = \sqrt{2/(\ell+1)}\sin[ij\pi/(\ell+1)]$ contain the physical energy orbitals. The number-preserving Fock lift $\Gamma(W)$ of $W = \Phi V^\dagger$ maps ρ_A to a state ρ_{al} whose probabilities are Eq. (4) in the physical-mode occupation basis. Its energy is E_{al}. The one-body trace rearrangement inequality proves that this is the minimum over number-preserving Gaussian orbital rotations.

For arbitrary subsystem unitaries, instead let $p_r^\downarrow$ be all $q(\mathbf{n})$ in descending order and $E_r^\uparrow$ all $E(\mathbf{n}) = \sum_j n_j \varepsilon_j$ in

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ascending order. Then

$$E_{\rm p} = \min_U \mathrm{Tr}(U\rho_A U^\dagger H_A) = \sum_{r=1}^{2^\ell} p_r^\downarrow E_r^\uparrow. \tag{5}$$

Indeed, the overlaps between eigenbases form a doubly stochastic matrix; the trace is a linear function of this matrix, minimized by the opposite ordering permutation, which is itself unitary. Degeneracies are retained. Define $\Delta W = E_{\rm al} - E_{\rm p} \geq 0$. If operations must commute with $\widehat{n}_A = \sum_j c_j^\dagger c_j$, one sorts within each number sector instead, obtaining $E_{\rm p}^{(n)}$ and $\Delta W_n = E_{\rm al} - E_{\rm p}^{(n)}$. Sector probabilities are neither discarded nor renormalized.

III. A CHARGE-CONSERVING COUNTEREXAMPLE AT $N = 12$

For $\ell = 6$, let $d_j^\dagger = \sum_i \Phi_{ij} c_i^\dagger$ and choose

$$|a\rangle = d_2^\dagger d_3^\dagger |0\rangle, \qquad |b\rangle = d_1^\dagger d_4^\dagger |0\rangle. \tag{6}$$

Both states contain two particles and have even parity, with the other modes empty. Their energies and aligned probabilities obey the strict rational enclosures

$$\begin{aligned} &0.335125603737 < \frac{E_b - E_a}{J} < 0.335125603738, \\ &2.002576633338 < 10^6(q_b - q_a) < 2.002576633339. \end{aligned} \tag{7}$$

Thus the higher-energy state has higher probability. These signs use the actual projector (2), not an asymptotic spectrum. For orientation, the descending occupations are approximately

$$\begin{aligned} (\nu_1, \ldots, \nu_6) \simeq (&0.999999987458,\ 0.999800520240,\\ &0.899541716786,\ 0.100458283214,\\ &0.000199479760,\ 0.000000012542). \end{aligned} \tag{8}$$

The approximate entries are not used to decide a sign.

Consider the exact rank-state swap

$$S = I - |a\rangle\langle a| - |b\rangle\langle b| + |a\rangle\langle b| + |b\rangle\langle a|. \tag{9}$$

It satisfies $S^\dagger = S$, $S^2 = I$, and commutes with both $\widehat{n}_A$ and $(-1)^{\widehat{n}_A}$. Its work gain from $\rho_{\rm al}$ is

$$g = \mathrm{Tr}[(\rho_{\rm al} - S\rho_{\rm al}S^\dagger)H_A] = (E_b - E_a)(q_b - q_a), \tag{10}$$

with

$$6.71114703278 \times 10^{-7} < g/J < 6.71114703279 \times 10^{-7}. \tag{11}$$

Consequently $U = S\Gamma(W)$ takes the original marginal to an energy below $E_{\rm al}$. Every density eigenvalue is preserved. The universal finite equality $E_{\rm p} = E_{\rm al}$ is therefore false, including when particle number and parity conservation are imposed. No environment operation or measurement enters this witness.

A quartic operation avoids projectors onto empty spectator modes. Set

$$K = d_1^\dagger d_4^\dagger d_3 d_2 + \text{h.c.}, \qquad U_K = \exp(-i\pi K/2). \tag{12}$$

After alignment it exchanges the same two four-mode patterns for every occupation of modes 5 and 6. Fermionic phases do not affect the energy. Spectator probabilities sum to one, giving

$$\begin{aligned} g_K = (&\varepsilon_1 + \varepsilon_4 - \varepsilon_2 - \varepsilon_3)\\ &\times [\nu_1(1-\nu_2)(1-\nu_3)\nu_4\\ &\quad - (1-\nu_1)\nu_2\nu_3(1-\nu_4)], \end{aligned} \tag{13}$$

and

$$6.71248612208 \times 10^{-7} < g_K/J < 6.71248612209 \times 10^{-7}. \tag{14}$$

This generator conserves number and parity. Its physical-mode interaction is not asserted to be spatially local, fast, or inexpensive.

IV. FIRST FAILING SIZE AND FINITE MAGNITUDE

The earlier allowed sizes $N = 4, 8$ are exactly passive after alignment. To see this, the staggered matrix $D_{jj} = (-1)^j$ anticommutes with the one-body hopping. With no zero mode, $DP_-D = I - P_-$ for the occupied projector. Restriction gives $D_A C_A D_A = I - C_A$, so occupations come in complementary pairs. Relative to the filled negative-energy local state, each pair has a hole and a particle excitation with equal positive energy cost and equal entanglement cost $\kappa = \log[\nu/(1-\nu)]$, $\nu > 1/2$. Probability is proportional to the exponential of minus the total entanglement cost.

For $N = 4$ there is one cost repeated twice. Increasing excitation count increases energy and decreases probability, proving passivity. For $N = 8$ the two energy costs, each repeated twice, are $J\alpha = J(1+\sqrt{5})/2$ and $J\beta = J(\sqrt{5}-1)/2$, with $\alpha > 2\beta$. Rational interval evaluation of Eq. (2) proves

$$\nu_1(1-\nu_2)^2 > (1-\nu_1)\nu_2^2, \tag{15}$$

hence $\kappa_1 > 2\kappa_2$. Excitation counts $(s_1, s_2) \in \{0,1,2\}^2$ are therefore lexicographically ordered by both physical and entanglement costs: increasing s_1 dominates every possible decrease in s_2, and at fixed s_1 both costs increase with s_2. States with the same counts have equal energy and probability. This proves $E_{\rm p} = E_{\rm al}$ at $N = 8$, rather than inferring equality from a small numerical residual. Thus $N = 12$ is the first failure in the sequence $N = 4m$.

Table I uses every Fock state at each size, totaling 5460 states over $N = 4, 8, 12, 16, 20, 24$. At $N = 12$, $E_{\rm al} \simeq -3.40404538383307J$ and $E_{\rm p} \simeq -3.40404674022927J$. The aligned bound energy is approximately $0.0899138J$, and its aligned ergotropy is $0.0365547J$. The correction

TABLE I. Complete finite-spectrum corrections as strict enclosures. The last column gives $10^6\Delta W/Q_{\rm al}$ approximately, with $Q_{\rm al} = E_{\rm al} - E_{A,0}$. Both corrections vanish exactly at $N = 4, 8$.

N	$10^6\Delta W/J$	$10^6\Delta W_n/J$	ppm
12	(1.356396207356, 1.356396207357)	(1.351749675589, 1.351749675590)	15.09
16	(1.871176478412, 1.871176478413)	(1.863882069329, 1.863882069330)	23.52
20	(2.029342670961, 2.029342670962)	(2.028803336503, 2.028803336504)	28.47
24	(2.003459793533, 2.003459793534)	(2.001606649667, 2.001606649668)	31.00

is about 15.09 ppm of the former and 37.11 ppm of the latter. It is an exact optimization distinction with a small quantitative effect.

V. ERROR CONTROL AND INTERPRETATION

The strict bounds are computer-assisted enclosures of the exact finite input. Every arithmetic operation rounds outward on rational intervals with denominator 10^{90}. Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ is enclosed using alternating series with 120 and 40 terms and next-term bounds. Angle reduction puts sines in $[0, \pi/2]$. A degree-199 Taylor polynomial, with its zero degree-200 coefficient, has remainder bounded by $2^{201}/201!$. This encloses Eq. (2) and all physical energies.

Numerical proposals for eigenvectors are only checked inputs. For every rational vector v and center μ, exact interval arithmetic verifies

$$\|(C_A - \mu I)v\|^2 < 10^{-90}\|v\|^2. \tag{16}$$

The Hermitian spectral decomposition then puts an eigenvalue within 10^{-45} of μ. All ℓ intervals are strictly disjoint and inside $(0, 1)$, so together they isolate the entire correlation spectrum. No assumed eigenvector orthogonality or floating eigensolver accuracy enters this argument.

Every product probability and subset energy is enclosed. Order statistics are monotone under componentwise bounds; sorting all lower endpoints and all upper endpoints therefore encloses the corresponding sorted exact values. Interval products and sums enclose Eq. (5), including negative energies and degeneracies, and the same procedure applies within number or parity blocks. A separate direct ground-state calculation in the 924-dimensional half-filled $N = 12$ sector, followed by an ordinary site-Fock partial trace, corroborates the marginal, orbital alignment, swap, and quartic energy traces to better than $2.2 \times 10^{-15}J$ for the two gain formulas. This floating cross-check is not the proof of the strict bounds. The exact checker and its input proposals are supplied as a reproducibility resource.

The physical correction is $W = W_{\rm al} + \Delta W$ and $Q = Q_{\rm al} - \Delta W$, where $W_{\rm al} = \mathrm{Tr}(\rho_A H_A) - E_{\rm al}$ and $Q = E_{\rm p} - E_{A,0}$. Occupation alignment remains valid with an explicit restriction to number-preserving Gaussian orbital rotations, or as an upper bound on the unrestricted passive energy. The additional work survives number and parity conservation. We establish no classification for $N > 24$, no thermodynamic error estimate, and no experimental or finite-time protocol. In particular, a finite failure of Eq. (34) does not determine the fate of the asymptotic entropy relation in Ref. [1].

a. Research tools. OpenAI Codex was used for the witness search, derivations, verification-code development and execution, literature comparison, and manuscript drafting. Exact arithmetic checks and separate numerical representations are described above; they do not constitute completed human scientific review.

b. Data availability. The accompanying reproducibility resource contains the exact checker, rational eigenvector proposals, complete certificate, and optional direct-density code and data.

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